

Lesson 6 Proving Universal and Existence Statements

Prove the following statements.

1. The product of two odd integers is odd.
2. For every integer x , if x is even, then x^2 is even.
3. For all integers x and y , if $x + y$ is odd, then $x - y$ is odd.
4. For all integers a, b, c , if $a|b$ and $b|c$, then $a|c$.
5. For all integers a, b, c , if $a|b$ and $a|c$, then $a|(b + c)$.
6. The difference between two consecutive squares is odd.

Prove or disprove:

7. If $6|ab$, then $6|a$ or $6|b$.
8. Let $x, y \in \mathbb{Z}$. If $x^2|y$, then $x|y$.
9. Let $x, y \in \mathbb{Z}$. If $x|y$ and $y|x$, then $x = y$.
10. For all integers a, b, c , if $ac = bc$, then $a = b$.
11. Let $a, b, c \in \mathbb{Z}$. If $a|bc$, then $a|b$ or $a|c$.
12. The sum of three consecutive integers is even.
13. Let $a, b, c \in \mathbb{Z}$. If $ab > ac$, then $b > c$.
14. Let a, b be integers. If $a < b$, then $a^2 < b^2$.
15. There are no integers a and b such that $7a + 11b = 1$.
16. There is an integer which is even and prime.
17. There exists a real number x such that $x^2 - 3x + 3 = 0$.
18. For all real numbers x , $|x| = |-x|$.
19. For all integers x , $x^2 \geq x$.
20. For all real numbers x , there is a natural number y such that $y > x$.